

Qn 1:

Title 1: Water Jug Problem (4-Gallon and 3-Gallon Jugs)

Agent Used: Simple Problem-Solving Agent

Reason: It searches for a sequence of valid actions to reach the goal state from the initial state.

Problem Statement:

The Water Jug Problem involves two jugs with capacities of 4 gallons and 3 gallons, both without measurement markings. The only allowed operations are filling a jug, emptying a jug, and pouring water from one jug to another. The challenge is to obtain exactly 2 gallons of water in the 4-gallon jug using these operations. The solution requires finding the correct sequence of steps while satisfying all the given conditions.

Assumptions

- The 4-gallon jug has a maximum capacity of 4 gallons.
- The 3-gallon jug has a maximum capacity of 3 gallons.
- Both jugs are initially empty.
- Water can be filled completely from the pump.
- Water can be poured from one jug to another.
- Water can be emptied onto the ground.
- No other measuring devices are available.

Qn 2:

Title 2: Mars Rover – Analyze and Explore the Samples

Agent Used: Goal-Based Agent

Reason: A Goal-Based Agent selects actions based on achieving specific goals, such as collecting samples, avoiding obstacles, and successfully completing the mission.

Problem Statement:

The Mars Rover is an autonomous robotic system designed to explore the surface of Mars and perform scientific research. It collects rock and soil samples, captures images, analyzes the environment, avoids obstacles, and transmits scientific data back to Earth. Since communication between Earth and Mars is delayed, the rover must make decisions independently without continuous human control. The objective is to analyze the Mars Rover as an intelligent agent by identifying its percepts, operating environment, possible actions, performance measures, and the most suitable agent architecture.

Qn 3:

Title 3: 8-Queens Problem

Agent Used: Simple Problem-Solving Agent (Search Agent)

Reason: The agent searches through different board configurations and finds a valid solution using the Backtracking search strategy.

Problem Statement:

The 8-Queens Problem is a classical Artificial Intelligence search problem. The objective is to place eight queens on an 8×8 chessboard so that no two queens attack each other. A queen can attack another queen if they are placed in the same row, column, or diagonal. The challenge is to find a valid arrangement of all eight queens by exploring different board configurations while satisfying all the given constraints.

Qn 4:

Title 4: OLA Cab Booking – Intelligent Agent

Agent Used: Goal-Based Agent

Reason: The Goal-Based Agent makes decisions to achieve the customer's goal of reaching the destination by selecting the most suitable available cab.

Problem Statement:

The OLA Cab Booking system is an intelligent transportation application that helps customers travel from one location to another. The customer enters the pickup location, destination, and preferred cab type, such as Mini, Micro, Sedan, Shared, or Prime. The system checks the availability of nearby cabs, estimates the fare, and confirms the booking. The objective is to provide a safe, convenient, and efficient travel service by selecting the most suitable cab for the customer.

Qn 5:

Title 5: Uniform Cost Search (UCS)

Agent Used: Simple Problem-Solving Agent

Reason: The Simple Problem-Solving Agent searches all possible paths and selects the one with the lowest cumulative cost to reach the destination.

Problem Statement:

A logistics company operates a network of warehouses connected by routes with different travel costs. Each route has a cost based on factors such as fuel consumption and travel time. The objective is to transport packages from the starting warehouse S to the destination warehouse G by finding the least-cost path. The Uniform Cost Search (UCS) algorithm is used to explore the weighted graph and identify the optimal path with the minimum cumulative cost.